BALA RAMESH C

SOFTWARE DEVELOPER

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PROFESSIONAL SUMMARY

Bachelor's degree in Electronics and Communication Engineering (ECE) with hands-on experience in full-stack web development. Proficient Full Stack Developer skilled in Python (FastAPI, Flask, Django), REST APIs, WebSockets, PostgreSQL, MySQL, Docker, React.js, Next.js, and OCR/OpenCV. Experienced in building scalable applications and collaborating with AI/ML teams to integrate AI tools and technologies into production-ready web systems.

EXPERIENCE

Junior Software Developer

RedBlox Technologies Private Limited, Pondicherry — Feb 2025–Jan 2026

- Developed and maintained web applications using **Next.js, Python (FastAPI)**, and databases including **PostgreSQL, MySQL, and Firebase**.
- Designed and implemented **JWT-based authentication, RESTful APIs, WebSockets, caching, and parallel processing** to support real-time, high-performance systems.
- Integrated an **online image editing platform** using **OpenCV** for real-time image manipulation.
- Built **reusable and responsive UI components** using **React.js, Tailwind CSS, and Bootstrap**, improving scalability and user experience.
- Implemented **OCR solutions** using **Tesseract, Doctr, and OpenCV** for text extraction, background removal, cropping, and shape detection.
- Enhanced **search performance and data retrieval** using **Typesense**.
- Collaborated with **cross-functional teams** to design, develop, and deploy **scalable software solutions**.
- Optimized **application performance, security, and maintainability** by following best practices in software development and system architecture.
- Worked closely with **AI/ML teams** to explore, evaluate, and integrate **AI tools and technologies** into web applications.

Office Administrator

Agaram Software Academy — Jan 2023- Jun 2024

- **Managed day-to-day office administration**, including documentation, scheduling, record maintenance, and professional correspondence.
- **Coordinated with students and faculty** to support enrollment processes, attendance tracking, and examination management.
- **Maintained and updated records, reports, and databases**, ensuring data accuracy, operational efficiency, and smooth workflow.
- **Supported administrative operations** by organizing files, handling inquiries, and assisting in process improvement initiatives.

SKILLS

Technologies:

Backend: Python, FastAPI, Django, Flask, REST APIs, WebSockets
Frontend: React.js (Redux), Next.js, JavaScript (ES6), HTML5, CSS3, Tailwind CSS, Bootstrap
Databases: PostgreSQL, MySQL, Firebase
Tools: Docker, GitHub, OpenAPI, Typesense
AI / Image Processing: OCR, OpenCV, Tesseract
Other Tools: Twilio, Figma, Ansoft HFSS

PROJECT

RedBloxAI

- Developed a scalable web platform using **FastAPI, Next.js, and PostgreSQL/MySQL**.
- Implemented **JWT-based authentication**, secure **RESTful APIs**, and **WebSockets** for real-time features.
- Integrated **OpenCV-based image processing** and **OCR pipelines** using **Tesseract**
- Optimized **search and indexing** with **Typesense**, significantly improving query performance.
- Enhanced overall system **response time, accuracy, and scalability** through performance tuning.

AI-Powered Hotel Operating System

- Designed and developed an AI-Powered Hotel Operating System using FastAPI and PostgreSQL, enabling efficient hotel booking, guest management, and reservation workflows.
- Implemented JWT-based authentication and role-based access control to securely manage hotel admins, staff, and guest users.
- Built a normalized PostgreSQL database schema for hotels, rooms, bookings, guests, and payments, ensuring data integrity and optimized query performance.
- Integrated Mailtrap for email services, enabling automated booking confirmations, cancellation notifications, and user verification emails in a safe testing environment.
- Developed RESTful APIs for hotel search, room availability, booking creation, and status management, with proper validation, error handling, and logging.

AI Content-Engine With React And Python Flask

- Built an AI-driven platform to automate email content generation and template management.
- Implemented **Gemini AI** to automatically generate dynamic email content based on user-provided inputs and prompts.
- Developed backend services using **Flask, SQLAlchemy, and MySQL** to handle AI requests, template storage, and email workflows.
- Built a responsive frontend using **React.js** with **Redux** for seamless user interaction and state management.
- Enabled direct email sending, automated content generation, and real-world productivity workflows.
- **Github Link:**https://github.com/BalaRamesh2411/AI-Content-engine
- **Reviewed By:**Mr.Saravana Kumar (Working at IBM)
- **Live Link:**http://ai-writer.agaram.academy/

CERTIFICATIONS

Full Stack Web Development

Completed 7-Month Full-Stack Web Development Course at Agaram Software Academy, Nagercovil

FRONTEND Development

Completed 6-Month Frontend Development Course at Login360,Chennai

EDUCATION

Bachelor of Engineering – Electronics and Communication Engineering

DMI Engineering College · 2022

ACHIEVEMENT

JOURNAL PUBLISHED :

Sustainable Diabetic Retinopathy Diagnosis System Using IoT:

- Designed an IoT-based healthcare system for remote monitoring using **Dexcom G4 Platinum sensors**.
- Developed a data analytics pipeline to generate predictive clinical insights for early detection of diabetic retinopathy.
- Achieved **99.58% accuracy**, **72.51% sensitivity**, and **99.83% specificity** in experimental evaluations.
- **Published in:** *International Research Journal of Multidisciplinary Technovation (IRJMT)*
  DOI: 10.34256/irjmtcon10